

Glycemic indices of five varieties of dates in healthy and diabetic subjects

.

Background This study was designed to determine the glycemic indices of five commonly used varieties of dates in healthy subjects and their effects on postprandial glucose excursions in individuals with type 2 diabetes mellitus. **Methods** Composition analysis was carried out for five types of dates (Tamer stage). The weights of the flesh of the dates equivalent to 50 g of available carbohydrates were calculated. The study subjects were thirteen healthy volunteers with a mean ($\pm$ SD) age of 40.2 ± 6.7 years and ten participants with type 2 diabetes mellitus (controlled on lifestyle measures and/or metformin) with a mean HbA1c ($\pm$ SD) of $6.6 \pm (0.7\%)$ and a mean age ($\pm$ SD) of 40.8 ± 5.7 years. Each subject was tested on eight separate days with 50 g of glucose (on 3 occasions) and 50 g equivalent of available carbohydrates from the 5 varieties of date (each on one occasion). Capillary glucose was measured in the healthy subjects at 0, 15, 30, 45, 60, 90 and 120 min and for the diabetics at 0, 30, 60, 90, 120, 150 and 180 min. The glycemic indices were determined as ratios of the incremental areas under the response curves for the dates compared to glucose. Statistical analyses were performed using the Mann-Whitney U test and repeated measures analysis of variance. **Results** Mean glycemic indices $\pm$ SEM of the dates for the healthy individuals were 54.0 ± 6.1 , 53.5 ± 8.6 , 46.3 ± 7.1 , 49.1 ± 3.6 and 55.1 ± 7.7 for Fara'd, Lulu, Bo ma'an, Dabbas and Khalas, respectively. Corresponding values for those with type 2 diabetes were very similar (46.1 ± 6.2 , 43.8 ± 7.7 , 51.8 ± 6.9 , 50.2 ± 3.9 and 53.0 ± 6.0). There were no statistically significant differences in the GIs between the control and the diabetic groups for the five types of dates, nor were there statistically significant differences among the dates' GIs ($df = 4$, $F = 0.365$, $p = 0.83$). **Conclusion** The results show low glycemic indices for the five types of dates included in the study and that their consumption by diabetic individuals does not result in significant postprandial glucose excursions. These findings point to the potential benefits of dates for diabetic subjects when used in a healthy balanced diet. Trial Registration Number ClinicalTrials.gov NCT01307904